AIRSTREET · PROBLEM STATEMENT

# AirStreet problem statement

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## DOCUMENT CONTROL

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| Project TRL | TRL 3 Proof of concept | | |
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## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.6 | 2026-10-24 | Amish Chadha | Kit 1.9.1 upgrade: users-and-needs and stakeholder tables (Tables 1 and 2) for the user requirements AST-URS-001; no other change |
| 0.5 | 2026-10-02 | Amish Chadha | First partner rule and first candidate to approach (AST-DEC-001, 2026-10-02) |
| 0.4 | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002) |
| 0.3 | 2026-09-25 | Amish Chadha | TRL 3 update; constraints and open questions reflect AST-DDR-001 and AST-CAL-001 |
| 0.2 | 2026-09-25 | Amish Chadha | Populate to TRL 2 (problem, users, context, constraints, prior work) |
| 0.1 | 2026-09-25 | Amish Chadha | Initial scaffold |

# AirStreet problem statement

Regulatory air monitors are few and far apart, missing the streets where people breathe traffic pollution.

## The problem

Traffic pollution varies sharply over short distances, but official monitoring is designed to describe whole cities. When Google Street View cars carrying reference-grade instruments drove every street in a 30 km² area of Oakland, California, nitrogen dioxide (NO2), nitric oxide and black carbon varied by up to 5 to 8 times within single city blocks (Apte et al., 2017). A single fixed station cannot show that pattern, and many cities have no station at all. Only 12 % of India's 4,041 census cities and towns have any air quality monitoring (CSE, 2023), and in 2020 OpenAQ found no public national government monitoring program in 13 populous countries, including Pakistan, Nigeria, Ethiopia, Kenya and Uganda (OpenAQ, 2020).

The health stakes are large. Ambient air pollution caused an estimated 4.2 million premature deaths worldwide in 2019 (WHO). NO2 comes mainly from burning fuel in cars, trucks, buses, power plants and off-road equipment (US EPA), so it is highest on busy streets, and more than 45 million people in the United States live within about 90 m (300 ft) of a major transportation facility (US EPA).

Low-cost sensors could fill the gap between stations, but raw readings are not trusted. Uncorrected PurpleAir particle sensors overestimated PM2.5 by about 40 % across the United States until a humidity-aware correction cut the error from 8 to 3 µg/m³ (Barkjohn et al., 2021). Electrochemical NO2 sensors respond to temperature, humidity and other gases, and laboratory calibration alone does not correct this; in a Pittsburgh study only field-calibrated models reached a mean absolute error of about 3.5 ppb for NO2 (Zimmerman et al., 2018).

AirStreet addresses one specific gap: an open, pole-mounted node that measures PM2.5 and NO2 at street level, with a calibration method stated up front, so a city, school or community group can build a neighborhood map that others can check.

## Users and context

Table 1. Users and needs (kit 1.8). The user requirements in AST-URS-001 are drawn from this table and Table 2.

| User type | Goals and needs | Typical use scenario | Pain points and challenges |
|---|---|---|---|
| Residents and community groups | See PM2.5 and NO2 on their own street, near schools and clinics, and whether it changes | A node on a pole outside a school, read on an open map | Raw low-cost sensor readings are not trusted; no reference station nearby; little money |
| Municipal environment and transport teams | Rank streets and judge interventions (school streets, bus corridors, freight routes) with a stated uncertainty | 10 to 100 nodes across a district beside at least one reference station | Need data often enough to see rush hours; need to know how good the numbers are; staff time at height is costly |
| Schools and universities | Teach and research exposure with raw data and a documented method; build and repair the node | A campus node built in a school workshop, data reprocessed by students | Closed products hide their method; commercial nodes are costly to buy and to license |
| Street furniture owners (utilities, city lighting) | Nothing that harms the pole, its wiring or its inspection; known load | Permission for a device on a light or signal pole | Drilling, added load and wind, access to mains in the pole, insurance |
| Researchers and data platforms | Raw signals and calibration versions, so data can be reprocessed and compared | Data pulled into an open platform such as OpenAQ | Data without calibration metadata cannot be reused |
| Installers and service crews | Fit and service quickly and safely from a platform or ladder | A pod exchange every 6 months beside traffic | Work at height next to traffic; small parts dropped from height |

Table 2. Stakeholders.

| Stakeholder group | Role or interest | Engagement needed |
|---|---|---|
| Reference station operator (for example the Texas Commission on Environmental Quality) | Hosts the 14-day and 6-monthly NO2 collocation and shares reference data | Agreement for a node beside a regulatory NO2 monitor at a street-level site (AST-DEC-001) |
| Pole owner (city lighting, utility, traffic department) | Permits the mount, confirms the load and wind on the pole, sets installation rules | Written permission, load check, inspection and insurance terms |
| City or air agency | Uses the maps; decides whether indicative data inform decisions | Co-design of the questions the network answers; data publication terms |
| Radio regulator and LoRaWAN network operator | Band rules (EU868 duty cycle, US915 dwell time) and fair use | Choice of band with the partner; private gateway or The Things Network |
| FieldNode (shared lab core) | Supplies the solar, battery and radio core; owns its mounting interface | Agreement to the adapter plate mount and the always-on 5 V port (AST-DDR-002) |
| CalRig (lab calibration rig) | Checks temperature, humidity and particle response before deployment | Chamber time for each pod |
| Open data platforms (for example OpenAQ) | Host calibrated hourly data with the calibration version | Data format and licence |

Table 3. Users and context (TRL 2 summary, kept for the record).

| User | Need | Context |
|---|---|---|
| Residents and community groups | See pollution on their own streets, near schools and clinics, and track whether it changes | Street poles on residential and arterial roads |
| Municipal environment and transport teams | Evaluate interventions such as low-traffic streets, school streets, bus corridors and freight routes | 10 to 100 nodes per district, alongside at least one reference station |
| Schools and universities | Teaching and research on exposure, with raw data and a documented method | Campus and nearby streets |
| Street furniture owners (utilities, city lighting) | A small, light device that does not interfere with the pole or its wiring | Street light and signal poles |
| Researchers and data platforms | Raw signals and calibration versions, so data can be reprocessed | Open data portals |

## Operating environment

- Mounted on a street light or signal pole, inlets 3.0 m above the sidewalk (AST-DDR-001 D7, decided by Amish on 2026-09-25), on the street side and within a few meters of the kerb.
- Outdoor, in sun, rain, dust and traffic spray; temperate to hot humid climates (about -10 to 50 °C, 10 to 100 % RH, estimate to be confirmed with partners). FieldNode is rated to 45 °C ambient, so sites hotter than that need FieldNode's sun shield (AST-DDR-002); the SPS30 particle sensor is recommended for 20 to 80 % RH, so the most humid sites remain at risk (AST-CAL-001).
- No mains power: runs on the FieldNode solar and battery core.
- Data over LoRaWAN: every 5 min through a private gateway (the lab's TwinKit gateway by default), or every 15 min on The Things Network, whose fair-use policy a 5 min interval would exceed (AST-DDR-002).

## Constraints

- Garage-buildable prototype. The $285 `budget_usd` covers the AirStreet sensor head; the FieldNode core is costed in FieldNode (AST-DDR-001 D1 and AST-DDR-002). The sensor head costs $285.00 at TRL 3 prices, within the budget with no margin (see 03-requirements.md).
- Built on FieldNode, the lab's shared outdoor core, with its power budget and sensor port pinout.
- Temperature, humidity and particle checks on CalRig. CalRig does not cover NO2, so NO2 must be calibrated by field collocation with a reference station.
- Levels only: no images, audio or personal data collected or sent.
- Mounting only with the pole owner's permission, without drilling the pole or opening its electrical access.

## Out of scope

- Regulatory compliance monitoring. AirStreet data are indicative and do not replace reference stations.
- Other pollutants (ozone, carbon monoxide, black carbon, ultrafine particles) in this version.
- Personal exposure devices and indoor use.
- Health advice or medical claims.

## Prior work

- Street-scale mapping with reference instruments on vehicles showed block-level variation (Apte et al., 2017).
- Community and city networks of low-cost sensors already operate, for example AirQo, which reports more than 400 PM2.5 sensors across 14 African countries (AirQo), and Breathe London, a sensing network developed with the Mayor of London (Breathe London).
- The US EPA published performance testing protocols for NO2, CO and SO2 sensors in February 2024, including field evaluation alongside regulatory monitors (US EPA).
- Humidity correction for optical particle sensors (Barkjohn et al., 2021) and field calibration of electrochemical gas sensors (Zimmerman et al., 2018) are established methods that AirStreet adopts rather than reinvents.

What is missing is an open, documented node design that pairs these sensors with a stated, repeatable calibration route and publishes raw signals.

## Open questions

- Which city and partner first, and which reference station can host a collocation? Decided on 2026-10-02 as a rule (AST-DEC-001): a city or air agency with a regulatory NO2 reference station at a street-level site and willing to share its data; first candidate to approach, the Texas Commission on Environmental Quality's Dallas-Fort Worth monitoring network with a local university or city partner. Nothing is agreed yet.
- Inlet height: 3.0 m is decided (AST-DDR-001 D7); partners may still ask for a lower inlet, closer to the breathing zone, which would reopen that decision.
- Is NO2 accuracy of about 4 to 5 ppb hourly, which cannot resolve the WHO annual guideline but can rank streets, useful enough for the first partner's questions? To be tested with users. Resolving the WHO guideline is no longer a requirement; it is kept as a research question (AST-DDR-002).

## User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [ ] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate load, distance, terrain and cost assumptions in the field
- [ ] Revise requirements (REQ) from findings before freezing the design